UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

NAAC GRADE A+ ACCREDITED UNIVERSITY

EXPERIMENT 6

Student Name: Jatin Tasoria **UID: 25MCC20054**

Branch: MCA-Cloud Computing **Section/Group: MCC 101-A**

Semester: 2nd **Date of Performance: 02/03/26**

Subject Name: Technical Training-I **Subject Code: 25CAP-652**

1. **Aim**

   To learn how to create, query, and manage views in SQL to simplify database queries and provide abstraction for end-users.

2. **Objective:**

   After completing this practical, the learner will be able to:

   - Understand the concept of data abstraction using views
   - Create simple and complex views
   - Restrict access to sensitive columns
   - Manage views using CREATE, ALTER, DROP
   - Apply views in real-world enterprise applications

3. **Practical / Experiment Steps**

   a. Start the system.
   b. Open pgAdmin.
   c. Create and select the database in which you want to perform the experiment.
   d. Establish connection to the database.
   e. Create simple and complex views
   f. Execute SELECT queries on views

4. Procedure of the Practical

1. Start the system and log in
2. Start the PostgreSQL service
3. Open PostgreSQL client (psql or pgAdmin)
4. Create a new database
5. Create the required table
6. Insert sample data into the table
7. Create views
8. Query views
9. Save the work for documentation

5. SQL Queries Used in the Experiment

- Simple View

```
CREATE VIEW active_employees AS
SELECT id, name, department
FROM employees
WHERE status = 'Active';
```

- View With Joins

```
CREATE VIEW employee_department_view AS
SELECT e.id, e.name, d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
```

- Summarization View

```
CREATE VIEW department_summary AS
SELECT department_id, COUNT(*) AS total_employees,
    AVG(salary) AS average_salary
FROM employees
GROUP BY department_id
```

6. Input / Output Analysis (I/O Analysis)

- Simple View:

	id integer	name character varying (50)	department_id integer
1	1	Rahul	1
2	2	Priya	2
3	4	Sneha	1

- View With Joins:

	id integer	name character varying (50)	department_name character varying (50)
1	1	Rahul	Information Technology
2	2	Priya	Human Resources
3	3	Amit	Finance

- Summarize View:

	department_id integer	total_employees bigint	average_salary numeric
2	4	1	35000.000000000000
3	2	1	38000.000000000000
4	1	2	47500.000000000000

7. Learning Outcome

- Mastery in creating and managing views
- Understanding abstraction and security concepts
- Ability to design views for payroll, HR, and reporting systems
- Real-world application of SQL abstraction layer